IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 30 April 2025, accepted 6 June 2025, date of publication 1 July 2025, date of current version 15 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3584886

RESEARCH ARTICLE

Estimating Import Lead Times Using Business Intelligence and Machine Learning Within the CRISP-DM Framework: A Case Study in Oil and Gas Services Industry

MOHAMED ANNIS SOUAMES[1], LARBI ABDERRAHMANE MOHAMMEDI[2], ISKANDER ZOUAGHI[2], ANGAPPA GUNASEKARAN[3], SAMIA BELDJOUDI[2], AND ABDERRAZAK LAGHOUAG[4]

[1]School of Built Environment, Engineering and Computing, Leeds Beckett University, LS1 3HE Leeds, U.K.
[2]Department of Industrial Engineering, Ècole Nationale Polytechnique, El Harrach 16200, Algeria
[3]School of Business Administration, Penn State Harrisburg, Middletown, PA 17057, USA
[4]Department of Business Administration, King Khalid University, Abha, Aseer 61421, Saudi Arabia

Corresponding author: Abderrazak Laghouag (alaghouag@kku.edu.sa)

This work was supported by the Deanship of Scientific Research at King Khalid University, Saudi Arabia, through Large Research Group Program under Grant RGP2/680/46.

ABSTRACT This study explores the estimation of import lead times in the oil and gas services industry by combining Business Intelligence (BI) and Machine Learning (ML) within the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework. Addressing the challenge of limited data availability, we employ data synthesis techniques alongside machine learning algorithms to model and predict lead times accurately across diverse import scenarios. The methodology begins with a dual perspective on the problem, encompassing both business and technical requirements, followed by systematic data collection, cleansing, and preparation. BI tools facilitate the visualization of key insights, allowing for enhanced data comprehension and performance tracking. Machine learning models are then applied to predict lead times under varying import conditions, optimizing decision-making in terms of transport mode, entry port, and customs handling. The final solution is deployed through an interactive chatbot interface, streamlining access to lead time estimates for import-export specialists. This integrated approach enables proactive lead time management, cost optimization, and greater operational efficiency within import processes, providing a strategic advantage in the industry.

INDEX TERMS Imports, lead time, business intelligence, machine learning, data synthesis, CRISP-DM, chatbot.

I. INTRODUCTION

The global oil and gas services industry operates within a complex network of international supply chains, where timely delivery of critical equipment and resources is crucial for maintaining smooth operations. Companies rely on efficient logistics to transport specialized machinery and components to operational bases, often situated in remote and challenging locations. The logistical complexity, coupled with diverse regulatory requirements, customs processes, and the coordination of multiple stakeholders, makes accurate import lead time estimation a critical factor for cost management and operational reliability. As De Treville et al. [1] and Ward and Zho [2] emphasized, precise lead time estimation plays a key role in reducing delays, controlling expenses, and improving responsiveness to evolving industry demands.

Achieving accurate lead time estimation, however, remains a considerable challenge. Traditional forecasting approaches

The associate editor coordinating the review of this manuscript and approving it for publication was Jenny Mahoney.

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often fail to adapt to the significant variability within global import processes, particularly in high-risk sectors like oil and gas. Chopra and Sodhi [3] underscored that such limitations are amplified in environments where delays in high-value shipments result in significant financial impacts. Similarly, Wang et al. [4] highlighted the need for adaptable models that can address these uncertainties. Compounding the issue, data scarcity and inconsistency further hinder accurate forecasts, particularly in remote regions common to oil and gas operations [5], [6].

To overcome these challenges, this study integrates Business Intelligence (BI) and Machine Learning (ML) within the CRISP-DM framework to develop a robust, data-driven model capable of predicting import lead times across varying logistical scenarios. Machine learning (ML) is commonly used in data mining to solve various problems. A popular framework for developing ML applications is the CRISP-DM [7]. Nodeh et al. [8] demonstrated the effectiveness of CRISP-DM in managing data mining projects, while Wahyudi and Arroufu [9] emphasized its value in complex environments. By leveraging the iterative structure of CRISP-DM, which includes business understanding, data understanding, data preparation, modeling, evaluation, and deployment, the study ensures a systematic approach to tackling lead time variability [10], [11].

The central research question addressed in this study is: *How can machine learning and data-driven techniques improve the accuracy of import lead time estimation in the oil and gas services industry?* To answer this, the study begins with a comprehensive analysis of business and technical requirements, followed by data collection, cleaning, and preparation. Machine learning techniques are then applied to generate predictive insights, addressing the constraints of limited and inconsistent data through data synthesis methods. As Sivarajah et al. [12] pointed out, integrating BI with machine learning enhances decision-making capabilities. Finally, the solution is deployed via an interactive chatbot interface, allowing users to obtain real-time lead time predictions. In a 2018 research by Arunachalam et al. [13], it was noted that such tools improve operational planning, offering actionable insights to manage supply chain complexities effectively. This integrated approach delivers a scalable and reliable solution for the unique challenges faced in the oil and gas sector.

The implications of this research are multidimensional. Theoretically, it advances the field of supply chain analytics by illustrating how machine learning can be applied to predict lead times in complex, data-constrained environments. Farshadfar et al. [14] highlight the value of such approaches for improving predictive capabilities in dynamic supply chains. Methodologically, the study demonstrates the CRISP-DM framework as a structured and practical guide for applying data science in the oil and gas industry, particularly when integrating Business Intelligence (BI) and Machine Learning (ML) for deeper insights [15], [16], [17]. Managerially, this research offers actionable insights for logistics and import-export managers, enabling proactive decision-making through data-driven strategies. Additionally, the chatbot interface developed in this study provides a user-friendly tool for accessing lead time estimates, significantly improving the efficiency and accuracy of import-related decision-making [18], [19].

The paper is organized into seven interconnected sections to provide a comprehensive analysis of lead time estimation in global supply chains, with a particular focus on the oil and gas services sector. The first section introduces the study's objectives, background, and structure, laying the groundwork for the subsequent sections. The second section highlights the pivotal role of lead time estimation in supply chain management, examining the limitations of traditional forecasting methods and the potential of BI and ML integration within the CRISP-DM framework.

The third section presents the research model, outlining the use of BI, ML, and the CRISP-DM methodology in supply chain analytics while detailing the specific machine learning models applied in this study. The fourth section describes the methodology, including the rationale for selecting CRISP-DM as the guiding framework, the unique challenges posed by the oil and gas industry, and the technical steps involved in data preparation, model development, and deployment.

The fifth section focuses on the case study, offering an overview of the industry's supply chain organization and the critical import processes that drive lead time estimation. The sixth section details the implementation of the CRISP-DM framework, explaining each phase from business understanding to deployment and presenting the resulting model performance and findings. The seventh section discusses the study's broader implications, providing theoretical, methodological, and managerial contributions while reflecting on its significance for supply chain operations. Finally, the paper concludes by summarizing the key contributions, acknowledging limitations, and suggesting directions for future research to further refine and generalize the proposed approach.

II. LITERATURE REVIEW

Effective lead time estimation is crucial for global supply chain efficiency, particularly in the gas & oil industry, where logistical challenges are complicated due to legal requirements and lengthy processes that span different remote locations. Recent progress in data analytics allows better visibility across the different supply chain stages, while the rapid development of machine learning techniques has given rise to a new set of techniques for estimating lead times in logistics and supply chains and hence allowing better optimization at a micro and macro level.

This section provides a literature review of the topic and is divided into three parts: the first part will analyse the impact of lead time estimation in global supply chains, the second part provides a review of the related works that integrated business intelligence (BI) and machine learning techniques

into supply chain optimization. Finally, the third part will address the use of CRISP-DM in literature as a data science framework applied for the specific use-case of supply chains.

A. THE CRITICAL ROLE OF LEAD TIME ESTIMATION AND LIMITATIONS OF TRADITIONAL APPROACHES IN GLOBAL SUPPLY CHAINS

Effective lead time management is critical for optimizing operations and maintaining competitiveness in global supply chains, particularly in high-stakes industries like oil and gas services. The prominence of lead time reduction gained traction with the rise of Just-in-Time (JIT) production systems in the 1950s, as discussed by De Treville et al. [1]. Companies in this sector navigate extensive, multinational supply chains that often involve remote and challenging locations, where accurate lead time estimation becomes essential for aligning logistics with project timelines, managing inventory, and ensuring service quality. Precise predictions allow firms to allocate resources efficiently while mitigating risks associated with disruptions such as regulatory delays or transport interruptions. Recent studies emphasize that accurate lead time estimation enhances operational efficiency and cost management, particularly in complex supply chains [20], [21].

However, achieving accurate lead time predictions is inherently challenging due to the variability and complexity of global logistics. In the oil and gas sector, shipments frequently involve specialized equipment that undergoes strict regulatory scrutiny, leading to unpredictable customs delays and clearance issues. Weijermars [22] and Basiri et al. [23] observe that logistical bottlenecks, infrastructure disparities, and varying documentation requirements across regions further complicate the estimation process. These factors render traditional deterministic models inadequate, as they fail to address the dynamic and variable nature of modern supply chains. Farahani et al. [24], and Escaith et al. [25] similarly note that lead time variability in international trade often stems from external influences, such as customs regulations and diverse transportation modes, which are difficult to capture with conventional forecasting techniques.

Traditional methods for lead time estimation, including regression analysis, time-series forecasting, and linear models, have been widely adopted for their simplicity and ease of implementation. These approaches rely on analyzing historical data to identify patterns, to extrapolate future lead times. While effective in stable environments, these models struggle to perform in industries like oil and gas, where unpredictability and variability are the norms. Framinan [26] and Ivanov [27] argue that deterministic models, though practical under controlled conditions, often fail in dynamic supply chain environments that frequently deviate from historical trends.

Moreover, traditional statistical models lack the flexibility to respond to sudden disruptions, such as customs hold-ups, route changes, or unexpected infrastructure delays. Enarsson [28] highlights that their reliance on static historical data limits adaptability, making them unsuitable for addressing new scenarios or unique regulatory challenges. To overcome these shortcomings, researchers have explored hybrid models that combine deterministic methods with stochastic or simulation-based approaches. For example, Nikolopoulou and Ierapetritou [29] shows that hybrid models can model the probability distributions of lead times to account for uncertainty. However, these models often fall short of capturing the dynamic, interdependent interactions within global supply chains.

Yang and Shami [30] further argue that while hybrid models improve adaptability through automated processes such as model selection and tuning, they remain limited in handling concept drift and variability, particularly in IoT-driven environments. Similarly, Sansana et al. [31] highlight that even robust hybrid models, which integrate mechanistic and data-driven frameworks, cannot match the real-time learning capabilities offered by advanced machine learning algorithms. In dynamic industrial settings, machine learning algorithms excel at adapting to disruptions, providing real-time predictions, and addressing the variability inherent in modern supply chains, making them a promising alternative for lead time estimation. Benlahreche et al. [82] also emphasize that integrating BPM with IoT can enhance real-time decision-making, ensuring more accurate predictions and improved operational efficiency in dynamic environments.

B. INTEGRATING BUSINESS INTELLIGENCE AND MACHINE LEARNING IN LEAD TIME ESTIMATION FOR THE OIL AND GAS SERVICES INDUSTRY

Business Intelligence (BI) and Machine Learning (ML) are driving transformative changes in supply chain analytics, enabling organizations to leverage data-driven insights for improved operational efficiency and decision-making. BI offers stakeholders real-time monitoring and visualization of critical performance metrics, including lead times, shipment statuses, and inventory levels. By converting raw data into actionable insights, BI tools provide transparency and control, essential for identifying and addressing bottlenecks within supply chains. Managing risks in supply chains, particularly in complex sectors like oil and gas, requires a structured approach to prioritize and mitigate threats. Business Performance Analytics (BPA), as an extension of Business Intelligence (BI), plays a vital role in identifying performance variables, uncovering risks, and revealing interdependencies within operations. Raffoni et al. [32] emphasized that effective BPA implementation requires addressing data quality, organizational competencies, and cultural shifts. Hatami-Marbini et al. [83] further developed a DEA-driven risk management framework that helps organizations prioritize risks, such as criminality and terrorist attacks, in the Nigerian oil supply chain. This integrated approach enhances decision-making and optimizes lead time prediction by addressing key challenges in risk management.

In the oil and gas services industry, where logistical operations are often complex due to specialized equipment and long-distance transportation, BI has emerged as a valuable tool for optimizing performance. Kocaoglu [33] highlights how BI tools empower companies in this sector to improve visibility and streamline operations, even in challenging logistics environments. By integrating BPA principles, organizations can further enhance their ability to diagnose performance issues holistically, encouraging cross-functional dialogue and proactive problem-solving.

While BI focuses on monitoring and visualization, ML introduces advanced predictive capabilities. By analyzing historical and real-time data, ML models can uncover hidden, non-linear patterns, enabling companies to anticipate disruptions and delays. Mohammadpoor and Torabi [34] emphasized that ML allows organizations to shift from static decision-making to adaptive strategies, particularly in industries where lead time variability depends on external factors like customs clearance processes, regulatory requirements, and transportation availability. Eissa [35] further observed that integrating ML within supply chain analytics significantly enhances operational resilience, making it possible to adapt to unexpected changes in dynamic logistics environments. In the oil and gas sector, this predictive capacity is critical, as delays in equipment shipments can have cascading effects on costs and timelines Al-Rbeawi [36]. Furthermore, as Bordeleau et al. [37] noted, the successful adoption of BI and analytics in organizations requires more than just technological resources; factors such as organizational learning and culture significantly influence the creation of business value. This perspective underscores the need for a holistic approach to BI and analytics integration, particularly in complex supply chains where adaptability and organizational alignment are critical for success.

With the rapid development of information technology, global competition has intensified, and the future of supply chain management (SCM) is poised for significant transformation, particularly with the integration of artificial intelligence (AI). Helo and Hao [38] highlighted that AI-driven approaches are reshaping key SCM domains such as planning, scheduling, optimization, and transportation. Their research underscores the value of AI-based business models in generating actionable insights and creating new opportunities for supply chain innovation. Fosso Wamba et al. [39] further emphasized the growing importance of AI applications in operations and SCM, noting their potential to offer practical benefits and to bridge gaps between industry practitioners and scholars. They called attention to the need for more research and dialogue to explore AI's transformative potential in this field.

Against this backdrop, machine learning techniques tailored to supply chain challenges have proven particularly effective for lead time prediction. Decision trees, for instance, are widely used for their interpretability, as they help identify significant variables such as transport delays and customs processing times. However, as Nodeh et al. [8] and Kapavarapu et al. [40] pointed out, decision trees tend to overfit in complex datasets. Random forests address this issue by averaging results across multiple decision trees, improving accuracy and effectively handling high-dimensional data in dynamic supply chains. Meanwhile, gradient boosting algorithms like XGBoost iteratively refine predictions by learning from prior errors, which makes them suitable for complex, non-linear relationships often encountered in supply chain processes [41]. Jin et al. [42] and Akbari and Do [43] demonstrated the effectiveness of Support Vector Machines (SVMs) in managing both linear and non-linear dependencies within interrelated variables, making them particularly relevant for predicting logistical metrics like lead times. Neural networks, although computationally intensive, provide a promising avenue for future applications when richer, large-scale datasets are available [44].

The synergy between BI and ML is increasingly seen as essential for addressing the challenges of modern supply chains. While BI tools provide the necessary visibility for monitoring KPIs and identifying disruptions, ML models enhance this capability by predicting delays and enabling organizations to act proactively. Kocaoglu [33] demonstrated that integrating BI dashboards with ML-driven predictive models has significantly improved lead time estimation, minimized delays, and optimized costs in the oil and gas industry. For instance, real-time analysis of shipment statuses, customs processes, and transport conditions allows companies to anticipate bottlenecks and take corrective measures before issues escalate.

Moreover, advanced enterprise systems such as Enterprise Resource Planning (ERP), Warehouse Management Systems (WMS), and Transportation Management Systems (TMS) further amplify the impact of BI and ML integration. Stefanovic et al. [45] emphasized how these systems offer a unified platform for data integration and operational control, enhancing transparency and enabling scenario-based planning. Zhu et al. [46] discussed the role of emerging technologies, including Digital Twins and Supply Chain Control Towers, in providing advanced analytics capabilities that align with the demands of global supply chains. Zouaghi and Laghouag [84] further highlighted how aligning key success factors (KSFs) with ERP implementation strategies can optimize system integration, thereby improving lead time estimation accuracy and overall supply chain performance.

The convergence of BI and ML represents a shift from reactive to proactive supply chain management. By improving lead time prediction, enhancing operational efficiency, and building resilience, this integration has proven particularly valuable in high-stakes industries like oil and gas. Moving forward, continued development and adoption of advanced analytics frameworks will enable companies to navigate the complexities of global logistics, ensuring adaptability, cost efficiency, and long-term competitiveness.

C. APPLYING THE CRISP-DM FRAMEWORK TO INTEGRATE BI AND ML IN SUPPLY CHAIN DATA SCIENCE PROJECTS

The CRISP-DM (Cross-Industry Standard Process for Data Mining) framework offers a structured and iterative approach for managing data science projects, comprising six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. Its widespread adoption in supply chain analytics is largely due to its ability to ensure alignment between technical implementation and business objectives, particularly in projects that integrate machine learning (ML) and Business Intelligence (BI). As Bokrantz et al. [47] and Harno et al. [48] emphasize, CRISP-DM is especially valuable for complex industries, where clear alignment at every phase of data analysis is essential for delivering strategic and operational value.

A notable application of CRISP-DM is its use in supporting strategic supply chain segmentation decisions, as highlighted by Kharlamov et al. [49]. Their research demonstrated how a CRISP-DM-based framework could help a multinational company transition from a single make-to-order strategy to a segmented supply chain strategy. By mining operational data, they identified key segmentation criteria, such as demand volume, demand variability, and delivery frequency, which enabled the development of tailored strategies for each segment. This example underscores CRISP-DM's capability to integrate data mining, BI, and ML to support strategic supply chain objectives.

Further demonstrating CRISP-DM's versatility, Jacobsen and Tan [50] explored its application in food safety management within a global fast-food franchise. By leveraging sensor-based monitoring systems and mining datasets containing over 4,000 alarm entries, they developed a transferable association rule model. This scalable model provided actionable insights and reliable visual information, enabling multiple managers to monitor their respective quality performance effectively. Their study highlights how CRISP-DM can contribute to digital transformation, offering empirical insights into the integration of data-driven systems for improved operational monitoring and decision-making.

One of the key strengths of CRISP-DM lies in its capacity to harmonize data mining (DM), BI, and ML across the analytics lifecycle. During the data understanding and preparation phases, DM techniques identify patterns and structure unorganized datasets, laying the groundwork for ML algorithms. These algorithms are then applied in the modeling phase to address specific supply chain challenges, such as lead time estimation or inventory optimization. BI tools play a complementary role during the evaluation and deployment phases by translating analytical outputs into actionable insights aligned with organizational goals. Bohanec et al. [51] note that this synergy creates analytics solutions that are both strategically relevant and operationally effective—critical in supply chains where decision-making must balance technological feasibility and managerial priorities.

The flexibility and comprehensiveness of CRISP-DM make it particularly suitable for integrating BI and ML into supply chain management. For instance, Gbededo et al. [52] demonstrated how structured methodologies like CRISP-DM help managers identify inefficiencies and address challenges in high-complexity environments. In the present study, CRISP-DM serves as the backbone for lead time estimation within the oil and gas sector. By leveraging data synthesis techniques to overcome data scarcity and employing ML models for prediction accuracy, this framework provides a practical solution for managing logistical complexities [8]. Similarly, Hamdani et al. [53] illustrated the iterative benefits of CRISP-DM in a logistics optimization project, showcasing its ability to refine predictive models continuously and navigate the socio-technical challenges associated with Big Data Analytics (BDA) implementation.

The work of Bohanec et al. [51] further validates the utility of CRISP-DM in integrating ML models with explanatory frameworks, facilitating organizational learning and decision-making. By employing action design research (ADR), their framework bridges the gap between technical models and user acceptance, ensuring that predictive analytics aligns with business needs. Their findings highlight the potential of CRISP-DM to enhance transparency and trust in ML-based solutions, a critical factor in high-stakes industries such as oil and gas.

In the current research, several machine learning models, decision trees, random forests, gradient boosted trees (GBT), support vector machines (SVMs), and linear regression, were evaluated for lead time prediction. These models were assessed using performance metrics such as mean absolute error (MAE) and root mean squared error (RMSE), with cross-validation ensuring their robustness and reliability. Amirian et al. [54] and Gharbi et al. [55] observed that integrating these models with BI dashboards provides logistics managers with intuitive, real-time platforms for decision-making, improving operational efficiency. Kumar and Sharma [85] further highlighted how machine learning models, such as random forests, can significantly enhance supply chain risk prediction, offering improved accuracy in forecasting lead times and mitigating potential disruptions, thus optimizing overall supply chain performance in the oil and gas industry.

Nevertheless, existing research reveals gaps in the integration of BI, ML, and CRISP-DM specifically for lead time estimation in the oil and gas sector. Most studies have focused on BI or ML independently, without offering a unified methodology for their application. Addressing this gap, the present study introduces a structured CRISP-DM-based framework that combines BI and ML to meet the sector's unique logistical and regulatory challenges. Nodeh et al. [8] stress the importance of CRISP-DM as a standard for achieving high-quality data analysis and predictive accuracy across projects.

By combining BI, ML, and CRISP-DM, this research develops a cohesive model that addresses the complexity of lead time estimation. BI enables data aggregation, visualization, and reporting, while ML enhances predictive capabilities, enabling logistics managers to proactively address potential disruptions. CRISP-DM ensures that these processes are organized into a clear methodology, with each phase aligning seamlessly with business objectives. Hamdani et al. [53] and Gbededo et al. [52] illustrate how such a structured approach enables actionable, data-driven decision-making.

The deployment of these solutions through user-friendly tools, such as chatbots and BI dashboards, further enhances their practical value by providing real-time insights to decision-makers. This study highlights the benefits of integrating structured methodologies with advanced analytics to tackle complex supply chain challenges. By bridging theoretical frameworks with practical applications, this comprehensive approach offers a robust model that can be applied not only in the oil and gas sector but also across industries grappling with similar logistical complexities.

III. METHODOLOGY

This study applies a structured and systematic approach to address the complexity of lead time estimation in the oil and gas services industry. The chosen methodology, CRISP-DM, provides a clear framework to guide each phase of data science and machine learning model development, ensuring that each step aligns with business objectives and operational needs. This section explains the rationale for selecting the CRISP-DM methodology, details the case selection process, addresses data challenges, and describes other critical methodological considerations.

A. METHODOLOGY SELECTION: CRISP-DM FRAMEWORK

There exist several frameworks that outline the steps for a proper study that relies on data and modeling similar to this paper. Some of these frameworks include: CRISP-DM, SEMMA and KDD, to name a few. CRISP-DM deals with the full cycle of data and modeling in a business context, it provides a sound business understanding. SEMMA is a framework focused on technical data manipulation and modeling while KDD is most suitable for extracting knowledge from structured data sources such as databases.

In the context of this study, The CRISP-DM framework was selected as it aligns data science techniques and machine learning modeling with business objectives and processes, an important aspect when studying complex supply chains. This business understanding and alignment is lacking in other frameworks such as SEMMA and KDD.

CRISP-DM has a structured iterative process composed of six phases, Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment, create a systematic pathway that progresses from problem definition to practical implementation. Wirth and Hipp [10] highlighted its effectiveness in managing complex projects, while Martínez-Plumed et al. [56] emphasized its adaptability to dynamic industries like oil and gas, where evolving project requirements demand ongoing model refinement to improve predictive accuracy.

Each phase of CRISP-DM is designed to ensure both technical precision and practical applicability of machine learning models. For instance, the Business Understanding phase focuses on aligning lead time estimation objectives with broader organizational goals. Following this, the Data Understanding and Preparation phases address critical data quality challenges, ensuring reliable inputs for the modeling process. Schröer et al. [11] noted the importance of this phase in delivering trustworthy results. The Evaluation phase provides a rigorous assessment of model performance, facilitating improvements where necessary, while the Deployment phase ensures that the predictive models are accessible to stakeholders. As Schröer et al. [57] discuss, this accessibility is essential for enhancing the model's operational value and ensuring its strategic alignment with business processes.

This structured methodology seamlessly integrates CRISP-DM with machine learning and BI, creating a robust approach for lead time estimation in the oil and gas services industry. By delivering actionable insights, it enhances operational efficiency and decision-making in complex supply chain environments.

Figure 1 illustrates the iterative nature of the CRISP-DM methodology applied in this study. The circular design underscores the continuous feedback loop between its six phases, with data positioned at the core of the process [58]. This central role of data ensures that the methodology remains aligned with business goals and operational needs, emphasizing adaptability and ongoing refinement in lead time estimation workflows within the oil and gas sector.

B. CASE STUDY: OIL AND GAS SERVICES

The oil and gas services industry was selected for this study due to its uniquely intricate supply chain and critical reliance on accurate lead time estimation. Delays in importing specialized equipment to remote or challenging locations can result in significant financial and operational repercussions. Unlike many other sectors, the oil and gas industry faces heightened complexities, including stringent regulatory requirements, extensive transportation routes, and the need to coordinate multiple stakeholders across diverse regions. Dempster et al. [59] emphasized the substantial logistical hurdles posed by these factors.

This challenging environment provides an ideal context for exploring the capabilities of machine learning (ML) and Business Intelligence (BI) in addressing operational inefficiencies. By leveraging the CRISP-DM framework, this study demonstrates the potential of ML-driven models to enhance lead time predictability and operational efficiency. Furthermore, the insights gained from this case study are not limited to oil and gas; as Goeken and Börner [60] suggested, similar supply chain challenges exist in high-stakes industries like

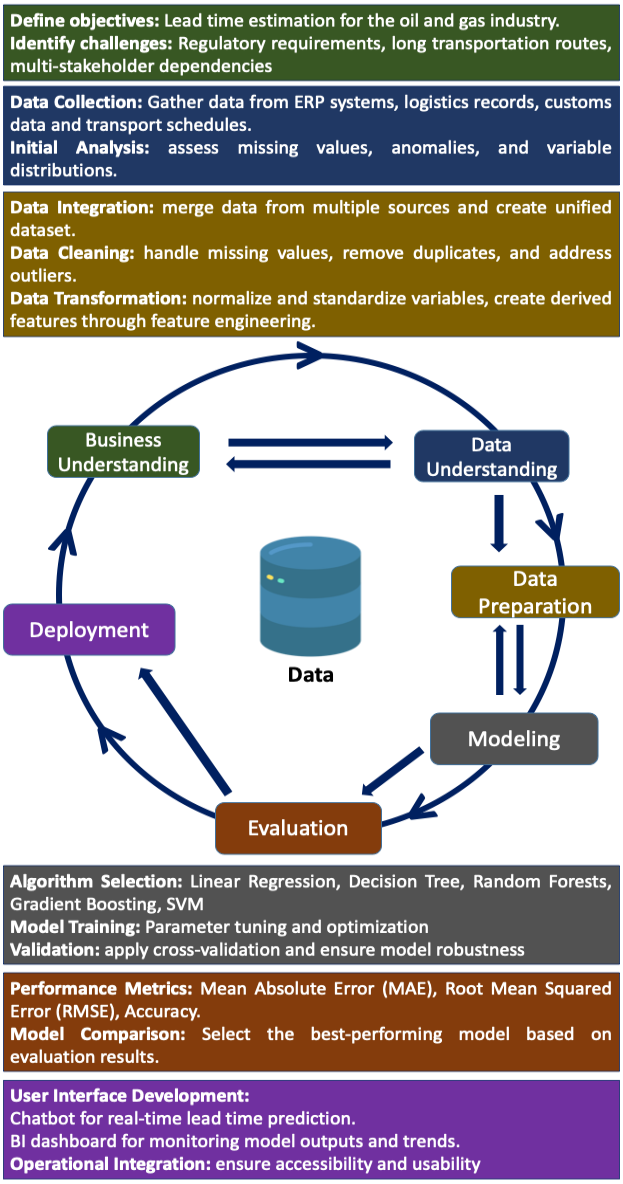

FIGURE 1. Iterative CRISP-DM Process for Lead Time Estimation.

aerospace and defense, making the findings applicable across other complex, globally interconnected sectors.

The global oilfield services market, projected to reach $134 billion in 2024, derives a substantial portion of its revenue from North America while experiencing rapid growth in regions like MENA, driven by rising demand for hydrocarbons and offshore discoveries requiring advanced technical services. According to Coherent Market Insights (2024), the market's structure, segmented by geography and function, enables providers to address regional demands through streamlined logistics, data-driven strategies, and adaptive resource management. Nevertheless, fluctuating oil prices and slower technological adoption in certain regions pose persistent challenges to efficient supply chain operations. The case study organization, an oilfield service provider, exemplifies these dynamics through its efficient supply chain structure, which includes three core functions: purchasing and supply, which manages supplier relationships and contracts to ensure procurement aligns with operational needs; global distribution, which oversees imports, exports, and inventory management to ensure timely resource availability; and domestic transport, which coordinates internal logistics to facilitate resource movement between operational bases.

The organization employs a standardized import-export framework comprising four stages. First, need validation and planning involve identifying and validating requirements in the ERP system, with requests fulfilled via stock transfers or purchase orders. Second, distribution ensures materials are transported internationally to HUB warehouses and subsequently distributed domestically. Third, customs clearance involves managing documentation, declarations, and duties to ensure regulatory compliance. Finally, final delivery oversees transport to operational bases, handles invoicing, and finalizes delivery documentation. This streamlined framework ensures that the organization minimizes delays and resource waste while maintaining operational efficiency in a challenging industry.

This study also examines the critical dates and lead times within the import process, as summarized in Table 1 and Table 2, respectively. The responsibilities across various stages are distributed among stakeholders such as business lines, Customs Clearance Agents (CCA), and customs authorities. This granular analysis enables a better understanding of the delays' root causes and identifies opportunities for optimizing the process. By integrating structured processes, predictive analytics, and real-time monitoring, the organization not only mitigates inefficiencies but also positions itself to adapt to the volatile and high-stakes demands of the oil and gas industry. These findings underscore the importance of advanced supply chain frameworks that leverage BI and ML to address sector-specific challenges.

C. DATA CHALLENGES, MODEL SELECTION, AND ETHICAL CONSIDERATIONS

Ensuring data quality and availability is fundamental to developing reliable predictive models, particularly in supply chains where datasets are often sparse or fragmented. This study tackles key challenges in lead time estimation within the oil and gas industry by employing advanced data processing techniques. To address data scarcity, synthesis methods were applied to generate synthetic data based on historical patterns, thus enriching datasets and improving the accuracy of model training. Kache and Seuring [61] highlighted the importance of such methods in mitigating data insufficiencies and enhancing model robustness. Furthermore, data integration consolidated information from diverse sources, including ERP systems, logistics records, customs data, and transport schedules. As emphasized by Obinwanne et al. [62], this unification ensures consistency, compatibility, and a comprehensive representation of the end-to-end import process.

TABLE 1. Dates that play a key part in the import-export processes and their definitions.

Key Date	Date Definition
CFD (Complete File Date)	Date the business line sends the complete declaration file to the CCA.
Declaration Date	Date the CCA declares the shipment to customs authorities.
VAT Submission Date	Date the VAT exemption request is submitted (for VAT regime shipments only).
VAT Recovery Date	Date the VAT exemption is received for the shipment (applicable for VAT regime shipments only).
Release Date	Date the release document is issued, allowing the shipment to exit the customs warehouse.
Delivery Date	Date the CCA transporter begins delivering the shipment from the customs warehouse to the operational base.
Reception Date	Date the shipment is received at the operational base.
Restitution Date	Date the container is returned to the entry port (applicable for sea freight shipments only).

TABLE 2. Main lead times for the import-export processes.

Lead Time	Lead Time Definition	Calculation Method	Responsible Entity
CFD Lead Time	Time taken by a business line to prepare declaration documents and send them to the CCA.	CFD Date - ATA Date	Business Line
Declaration Lead Time	Time the CCA takes to declare the shipment to customs authorities.	Declaration Date - CFD Date	Customs Clearance Agent (CCA)
VAT Declaration Lead Time	Time for the CCA to submit a VAT exemption request (for VAT regimes).	VAT Submission Date - Declaration Date	CCA
VAT Handling Lead Time	Time required for customs to process the VAT exemption request.	VAT Recovery Date - VAT Submission Date	Customs Authorities
Release Lead Time	Time to release a shipment from a customs warehouse, depending on the customs regime (FD or VAT).	Regime FD: Release Date - Declaration Date Regime VAT: Release Date - VAT Recovery Date	CCA
Transport Lead Time	Time required to transport the shipment from the entry port to the operational base.	Reception Date - Delivery Date	CCA/Transporter
Restitution Lead Time	Time taken to return the container from the operational base to the entry port (for sea freight only).	Restitution Date - Reception Date	CCA

In the data cleaning and transformation phase, issues such as missing values, outliers, and inconsistencies were resolved through standardized procedures. Variables were normalized, gaps were imputed, and scaling was applied to prevent any single variable from disproportionately influencing the model results [63]. To further enhance predictive power, feature engineering techniques were used to derive new variables—such as shipment weight or transport route complexity—while correlation analysis and feature importance ranking identified the most relevant predictors. These approaches, discussed by Rane et al. [64], ensured that the models captured critical relationships within the data.

To achieve an optimal balance between accuracy, flexibility, and interpretability, a variety of machine learning models were explored. Linear regression, praised by Chudik et al. [65] for its simplicity and ability to capture linear trends, served as a baseline for identifying initial patterns. For more complex relationships, decision trees were employed due to their interpretability, as they segment data into logical branches based on feature distributions. Nodeh et al. [8] and Vairagade et al. [66] further noted that random forests, by combining multiple decision trees, mitigate overfitting and enhance accuracy through ensemble learning. For non-linear relationships, gradient boosting models such as XGBoost were particularly effective, with Zhong et al. [67] highlighting their strength in minimizing prediction errors, making them well-suited for dynamic supply chain forecasting tasks. Support Vector Machines (SVMs), recognized for their ability to model intricate decision boundaries, were also tested. However, Akbari and Do [43] and Li [68] acknowledged the computational challenges of SVMs, particularly when dealing with large datasets.

The performance of all models was rigorously evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) as key metrics. To ensure model robustness and generalizability, cross-validation techniques were applied across multiple test scenarios, as advocated by Lim et al. [69]. This comprehensive evaluation ensured that the selected

models reliably addressed the variability inherent in lead time estimation.

Ethical and practical considerations were integral to this study's approach. Data privacy and security were prioritized through anonymization and secure storage of sensitive logistics data. Additionally, the predictive models were positioned as decision support tools, intended to complement human judgment rather than replace it. This perspective aligns with [70] assertion that responsible use of analytics ensures practical and ethical implementation, particularly in critical industries like oil and gas.

This integrative approach demonstrates the importance of combining advanced data science techniques with ethical considerations to deliver practical solutions for complex supply chain challenges. By addressing data limitations, improving model accuracy, and fostering responsible analytics, this study offers a robust framework for enhancing operational efficiency and decision-making in the oil and gas industry.

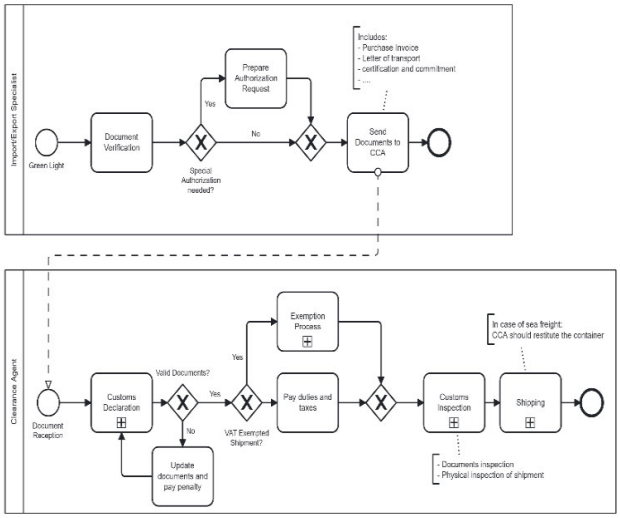

FIGURE 2. BPMN model of the import process: Collaboration between clearance agent and I/E Specialist.

IV. CRISP-DM DEPLOYMENT AND RESULTS OVERVIEW

The deployment phase in the CRISP-DM framework emphasizes making predictive models practical and accessible for operational use. In this study, a chatbot interface has been developed to provide logistics managers and import coordinators with real-time lead time estimates. Acting as a user-friendly tool, the chatbot delivers actionable insights without requiring technical expertise, thereby enhancing accessibility and facilitating informed, timely decision-making [11].

Complementing the chatbot, a BI dashboard is integrated to display model outputs and key performance metrics. This dashboard offers an interactive platform for monitoring predicted lead times, identifying factors contributing to delays, and analyzing shipment trends. By combining BI with machine learning outputs, the dashboard and chatbot work together to ensure predictions are both transparent and actionable [71].

Following the CRISP-DM methodology, this deployment phase addresses challenges within the import process, particularly lead time variability and its operational impacts. The study begins with a clear problem definition during the Business Understanding phase, identifying critical issues such as customs clearance delays and inconsistent lead times that increase costs and disrupt planning. The primary objective is to develop a predictive model that enhances lead time accuracy, enabling proactive resource allocation and reducing storage and demurrage expenses.

In the Data Understanding and Preparation phases, relevant data sources—including shipment details, customs processing times, and transport schedules—are identified and prepared through techniques such as missing value imputation, data standardization, and feature extraction. Machine learning algorithms like random forests and gradient boosted trees (GBT) are employed during the Modeling phase to build robust predictive models. Finally, in the Deployment phase, these models are seamlessly integrated into logistics workflows via a chatbot and BI dashboard, providing real-time insights that anticipate delays and support cost-efficient, data-driven decision-making across the import process.

A. BUSINESS UNDERSTANDING

To effectively address the challenges within the import process, it is vital to first understand its structure and the stakeholders involved. The analysis began with Business Process Model and Notation (BPMN) 2.0, which provided a clear visual representation of national lead times, emphasizing domestic activities and customs clearance stages. This mapping revealed crucial interactions among stakeholders and potential areas for inefficiencies.

To complement this, a Power BI dashboard was developed, enabling detailed monitoring and analysis of key aspects of the import process. The dashboard's primary objectives were to analyze shipment volumes by mode of transport and customs brokers, breakdown lead times across parameters such as import regime and business line, evaluate costs related to storage and customs clearance, and assess the performance of customs brokers. A dimensional modeling approach was implemented, structuring data into dimension and fact tables for efficient analysis.

Despite initial limitations in data sources—such as incomplete shipment details in SAP ERP and insufficient performance tracking in customs broker reports—the Clearance Portal and Quest were chosen as reliable sources, providing essential shipment information and broker performance evaluations.

The Power BI dashboard was structured into four main pages:

1. **Import Process Overview**: Highlighted shipments by import stages and identified bottlenecks.

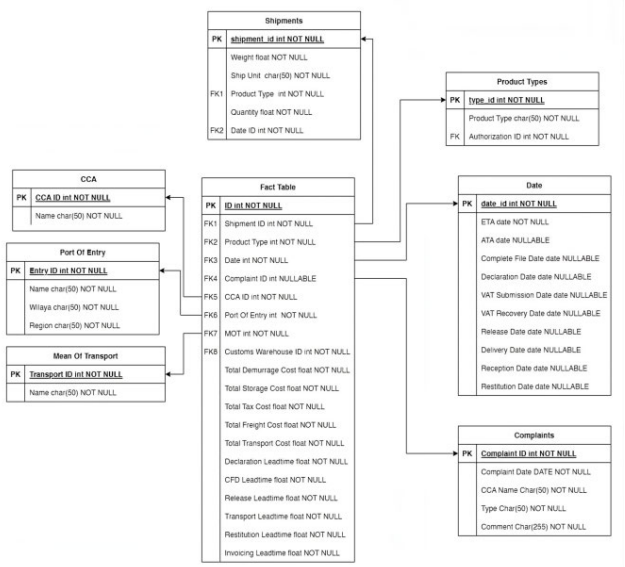

FIGURE 3. Proposed dimensional modeling for analyzing the import process.

2. **Lead Time Analysis**: Pinpointed delays by examining lead times across brokers, customs offices, and product lines.
3. **Cost Analysis**: Compared storage and customs-related costs against annual targets.
4. **Customs Broker Performance**: Assessed broker efficiency, compliance, and reliability for decision-making.

Key findings revealed that in 2022, a single customs clearance agent (CCA) managed 60% of shipments, with sea freight representing only 20% of shipments but contributing to 73% of storage costs. Delays in declaration lead times were frequent, and inefficiencies in MI entity processes particularly banking delays causing a 39-day documentation preparation period drove storage and demurrage costs above targets. Overall broker performance scores reached 87.97%, though shipping performance was significantly lower at 54.91% during the first half of 2022. While penalty clauses were included in broker contracts, they were inconsistently applied due to time constraints and limited resources within the imports team (manual review of contracts and cost calculations was often skipped), leading to case-by-case negotiations without proper penalty assessments.

To resolve this, we introduced a credit note system that automates monthly penalty calculations using Python scripts and data from the Clearance Portal. This system applies predefined rules based on delay types (e.g., airport-related delays triggering a 5% penalty on brokerage costs), ensuring consistent enforcement aligned with contractual terms and reducing the manual work.

Additionally, the study identified frequent requests for lead time and cost estimates, which were challenging due to the complexity of variables like entry ports and transport modes. To streamline this, a machine learning model was proposed to estimate lead times, simplifying cost estimation processes. Following the CRISP-DM framework,

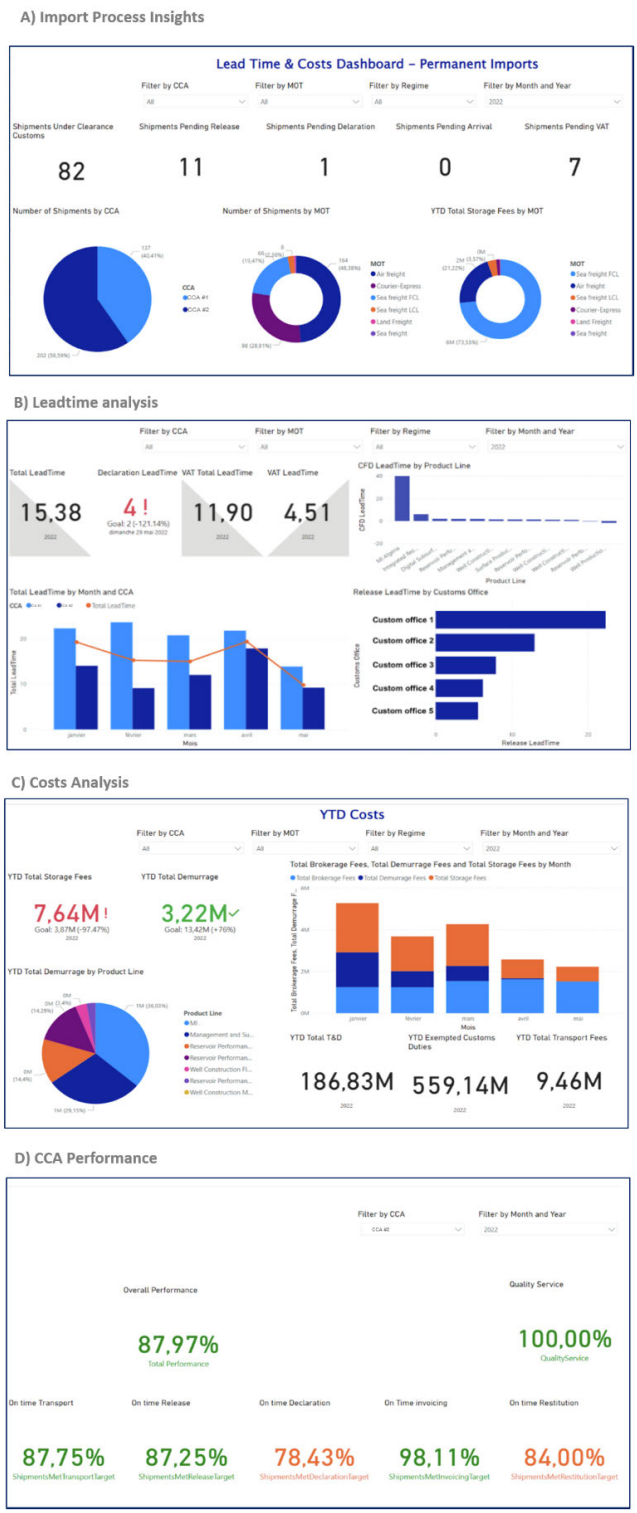

FIGURE 4. Power BI dashboard for import process analysis and optimization including Overview, lead time analysis, Costs and CCA performance.

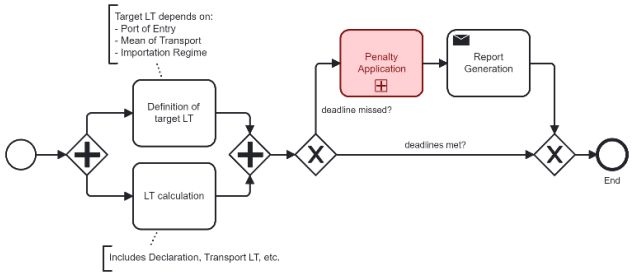

FIGURE 5. Proposed solution process for penalty application and calculation.

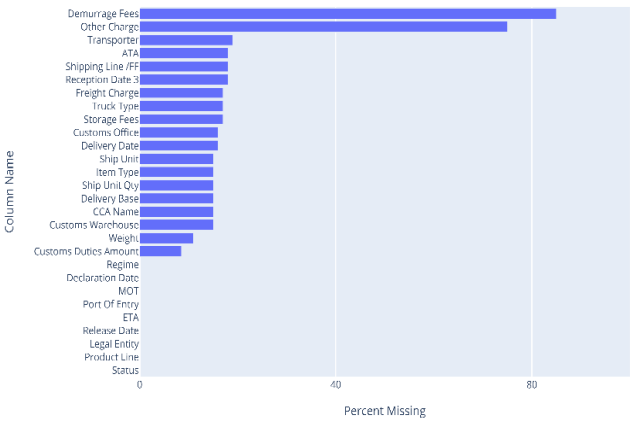

FIGURE 6. Percentage (%) of missing values in each selected column, providing a visual summary of data completeness across essential fields.

this solution emphasizes systematic data preparation, robust model development, and practical deployment to improve decision-making in the import process.

B. DATA UNDERSTANDING

To effectively address the complexities of the import process, the study began with a comprehensive exploratory data analysis (EDA) of the selected data source, the Clearance Portal. This platform, integral to the import-export department, was analyzed to identify data inconsistencies and ensure proper cleaning and preparation for subsequent modeling steps. The dataset revealed challenges, particularly with incomplete entries before 2022, as the platform had only been fully integrated by late 2021.

The analysis first assessed missing data in each column using the Pandas library. Figure 6 visually summarizes the percentage of missing values, revealing that critical columns, such as "MOT" (Mean of Transport), "Status," "Regime," and "Port of Entry," are complete. However, approximately 17% of data was missing in other columns, with cost-related fields showing over 70% missing values. Particularly crucial date columns, such as "ATA," "Declaration Date," and "Reception Date," were found to have about 17% missing data. To ensure accurate lead time computations, rows missing these key dates were excluded.

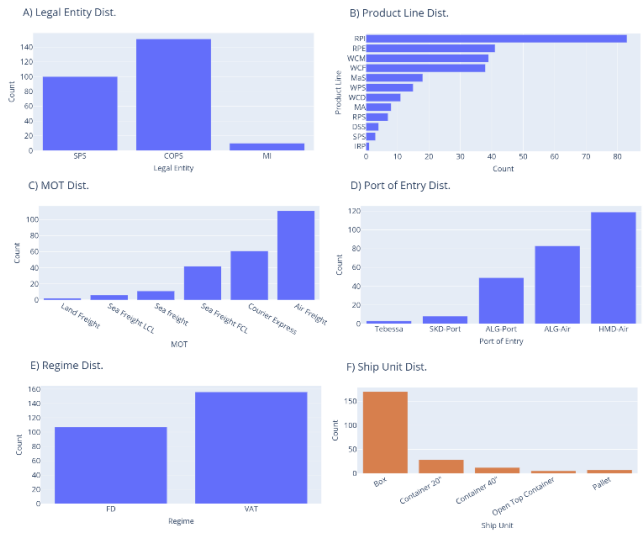

FIGURE 7. A) Distribution of Legal Entity, B) Distribution of Product Line, C) Distribution of Mean of Transport (MOT), D) Distribution of Port of Entry, E) Distribution of Regime, F) Distribution of Ship Unit.

Imbalances in categorical data were also evident. Figure 7 illustrates that the MI Entity accounts for less than 5% of total shipments, while 50% of shipments are transported via air freight, and 16% use sea freight (FCL). This underscores a significant concentration of shipments in specific categories.

Further analysis of shipment weights indicated that most shipments weighed below 2000 kg, with the "Box" unit representing 67% of all shipments. This aligns with the predominantly light-weight nature of shipments.

Lead time analysis, shown in Figures 8 and 9, identified "Release lead time" as the longest, averaging 7.6 days and acting as a primary bottleneck. Outliers in the "CFD lead time" reached up to 31 days. The distribution of lead times was also evaluated, revealing non-normal distributions for "CFD lead time" and significant right skewness in other lead times, indicating variability.

To determine the factors influencing lead time variability, a correlation analysis was conducted. Classical ANOVA tests were initially considered but deemed unsuitable due to unmet conditions of normality and homoscedasticity. The Shapiro-Wilk test (Table 3A) confirmed that lead times do not follow a normal distribution (p-values $< \alpha$). Additionally, Levene's test (Table 3B) highlighted unequal variances, such as those between "CCA Name" and "Declaration lead time" (p-value < 0.05). Consequently, we need to use a non-parametric test. Several non-parametric tests exist such as Friedman's test, Mann-Whitney U and Kriskal-Wallis, we opted for the Kriskal-Wallis test which is an extension of the Mann-Whitney U test used to compare three or more independent groups. It is worth noting that in this study, all the groups are independent and therefore the Friedman test is not as suitable as the Kruskal-Wallis/Mann-Whitney U statistical tests.

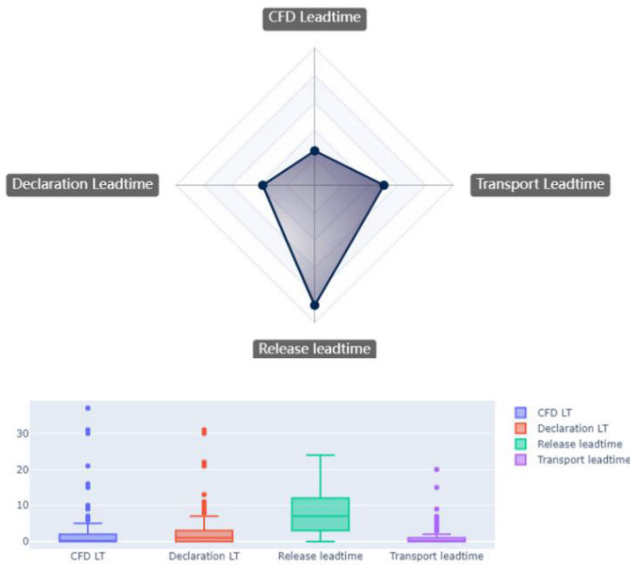

FIGURE 8. Analysis of the distribution and proportion of import lead times.

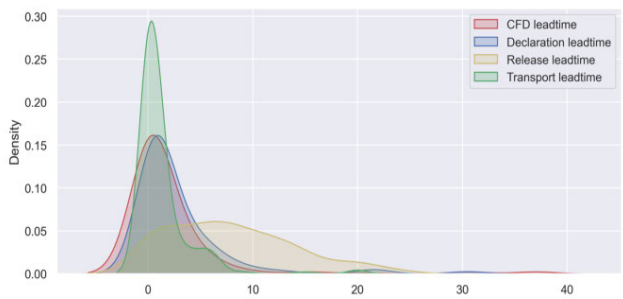

FIGURE 9. Estimated distribution of various lead times.

TABLE 3. A) Shapiro-Wilk Test on Various Lead Times. B) Levene's Test on Lead Times with "CCA Name," "MOT" (Mode of Transport), and "Regime".

A	
Lead Time	P-value
CFD	1.54×10^{-24}
Declaration	2.30×10^{-21}
Release	1.19×10^{-6}
Transport	8.14×10^{-23}

B			
Column / Lead Time	CCA Name	MOT	Regime
Declaration	1.43×10^{-4}	2.12×10^{-1}	9.83×10^{-1}
Release	3.12×10^{-1}	4.07×10^{-2}	7.38×10^{-2}
Transport	4.92×10^{-2}	1.49×10^{-1}	9.40×10^{-1}

The Kruskal Wallis test helped in identifying the top five variables most correlated with lead times, summarized in Table 4, based on η^2 statistics derived from the H statistic.

C. DATA PREPARATION

The data preparation process for this study involved cleaning and augmenting the dataset to ensure its suitability for machine learning modeling in the context of lead time estimation. The original dataset included 180 shipments after excluding incomplete records, focusing on key dimensions such as "CFD Lead Time," "Declaration Lead Time," "Release Lead Time," and "Transport Lead Time." The dataset includes also these dimensions mentioned in Table 5.

The limited dataset posed challenges, including risks of overfitting and reduced predictive accuracy. To address these issues, two data augmentation methods were used: simulation-based oversampling approach and a neural network-based approach through the Tabular Variational Auto-Encoder model (TVAE). These two techniques allow

TABLE 4. Top 5 Dimensions (Columns) most correlated with lead times according to the Kruskal-wallis test.

(a) For CFD Lead Time	
Source	η^2
Product Line	0.111807
Legal Entity	0.091130
Customs Office	0.052979
Port Of Entry	0.029962
Shipping Line	0.029704

(b) For Declaration Lead Time	
Source	η^2
CCA Name	0.224742
Transporter	0.187482
Shipping Line	0.102934
Legal Entity	0.066495
MOT	0.063672

(c) For Release Lead Time	
Source	η^2
Regime	0.281433
Transporter	0.267920
Shipping Line	0.199596
CCA Name	0.096268
MOT	0.094842

(d) For Transport Lead Time	
Source	η^2
MOT	0.106028
Port Of Entry	0.078817
Product Line	0.071038
Shipping Line	0.063303
Transporter	0.055862

TABLE 5. Dimensions used for machine learning models.

Dimension	Description
Product Line	The business unit that requested the shipment
Status	The status of the shipment
Legal Entity	The legal entity that requested the shipment
Weight	The weight of the shipment in kg
Ship Unit Qty	The quantity requested
Ship Unit	The logistics unit (container, pallet, etc.)
Product Type	The type of shipment (chemical, explosive, etc.)
Port Of Entry	The entry region of the shipment
MOT	The mode of transport
Regime	The import regime (exempt or not)
Service Level	The urgency of the shipment (urgent or not)
Delivery Base	The final destination of the shipment
Customs Office	The office responsible for customs clearance
Customs Warehouse	The customs warehouse
Forwarder	The name of the forwarder
Carrier	The name of the carrier of the shipment

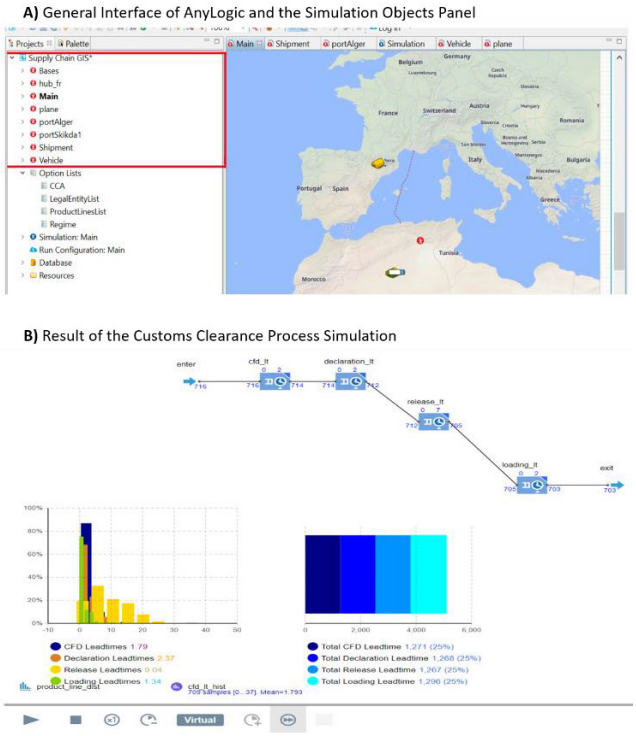

FIGURE 10. A) General Interface of AnyLogic and the Simulation Objects Panel (Top), B) Customs Clearance Process Simulation (Bottom).

generating synthetic data while preserving the original data's distribution and integrity.

For simulation-based oversampling, we used AnyLogic software to model the customs clearance process, focusing on major ports such as Algiers and Skikda. Simulation agents were created based on empirical distributions from operational data, respecting real-world constraints for example, shipments exceeding certain weight thresholds were assigned containerized shipping units and sea freight transport modes, while specific ports of entry (e.g., Algiers, Skikda, HMD Airport) were selected based on the chosen transport mode.

Generating data for supply chains is a complex task as the nature of the supply chain itself is composed of several components with different behaviours. To address this fragmented nature and ensure reliability and consistency of the augmented data, particularly for process time and lead time variables, we used empirical distributions for key lead times including "CFD lead time," "Declaration lead time," "Release lead time," and "Transport lead time" derived from the original dataset through a Python script. These were implemented in AnyLogic using the CustomDistribution function, supported by custom Java methods that dynamically calculated delays for each stage of the clearance process.

The resulting simulation generated approximately 700 synthetic shipments records in under 10 minutes, providing a significant increase to the dataset volume while reflecting realistic process delays and interactions.

Additionally, The TVAE technique was applied to learn the distribution and relationships within the original dataset, generating approximately 500 new rows of synthetic data. Compared to the previously discussed simulation method, TVAE offered advantages in speed, simplicity, and its ability to capture variable correlations automatically.

To ensure that the synthetic data adhered to real-world dependencies, a set of logical constraints was implemented using the **sdv.constraints** module in Python. These included:

- **Fixed combinations:**to preserve valid groupings of categorical fields such as:
  - 'CCA Name', 'Transporter', 'Transport LT'
  - 'MOT', 'Port of Entry', 'Delivery Base', 'Customs Warehouse', 'Customs Office', 'Weight', 'Ship Unit', 'Ship Unit Qty'
- **Between constraint:** on 'Full delay' column, limiting values between 1 and 50 to reflect realistic delay durations.
- **Positive constraint:** on lead time related fields such as 'CFD LT, 'Declaration LT', etc. To ensure all durations remained non-negative.

These constraints were enforced using either transformation of rejection sampling strategies, depending on the nature of the constraint, which enhanced the reliability and usability of the augmented dataset.

Data evaluation metrics, including Kullback-Leibler Divergence (DKL), confirmed that TVAE-generated data aligned better with the original dataset, outperforming simulation-based data in terms of distribution accuracy.

For the ease of evaluating the dataset using DKL and training the estimation ML models, we create 3 different datasets based on the original data and the generated synthetic data, denoted as follows:

1. **D1**: Contains only the original data, this is the original dataset which contains data for 180 shipments only.
2. **D2**: Combines the 180 original data points and the 700 synthetic shipments data generated through the simulation method, the resulting dataset D2 has a size of 880 shipment data.
3. **D3**: Combines the 180 original data points and the full 500 data points generated through the TVAE model, the resulting dataset D3 has a size of 680 shipment data.

Therefore, in terms of size D2 is slightly larger than D3, while both being over three times larger than the original dataset D1.

As a final processing step, the skewness of the weight variable was reduced through a logarithmic transformation, while the nominal features were encoded using mean target encoding since all the employed predictive models expect numerical features.

D. MODELING

1) LEAD TIME ESTIMATION

Baseline Model and Error Metric

To assess model performance, we define a baseline model, which employees at the case study company mentally use to estimate lead times. This baseline model simply estimates

TABLE 6. MAE metric for CFD, Declaration, Release and Transport lead times across different models.

Lead Time	CFD			Declaration			Release			Transport		
Dataset	**D1**	**D2**	**D3**	**D1**	**D2**	**D3**	**D1**	**D2**	**D3**	**D1**	**D2**	**D3**
Linear Reg.	1.71	0.89	0.96	2.43	1.22	1.16	3.32	4.26	**3.05**	1.39	1.09	0.68
Decision Tree	1.95	1.00	1.10	2.30	1.27	1.21	3.71	5.38	4.25	1.35	1.43	0.63
Random Forest	1.92	0.91	1.03	2.28	1.25	1.19	3.90	4.24	3.22	1.13	1.09	0.63
SVM	1.94	**0.75**	0.85	2.00	1.12	**1.04**	4.33	4.56	3.40	1.11	0.97	**0.54**
GBT	1.86	0.92	1.04	2.32	1.26	1.20	4.33	4.27	3.18	1.26	1.16	0.67
Baseline	2.47	2.47	2.47	2.53	2.53	2.53	4.83	4.83	4.83	1.43	1.43	1.43

lead times by averaging previous lead times. For each lead time, we will evaluate the performance of these models on datasets D1, D2, and D3, using the Mean Absolute Error (MAE) as our error metric:

$$MAE = \frac{1}{n}\sum_{i=0}^{n}\left|\hat{y}_i - y_i\right| \tag{1}$$

For the baseline error, we replace $\hat{y}_i$ with the mean $\bar{y}_i$, resulting in:

$$MAE_{ref} = \frac{1}{n}\sum_{i=0}^{n}|\bar{y}_i - y_i| \tag{2}$$

This error metric is easy to interpret since it has the same unit as lead times (days). We will then compare each model's MAE against the baseline error and select the model with the lowest MAE as the best-performing model.

2) IMPORT COST ESTIMATION

Having estimated the various lead times related to the import process, we now shift our focus to estimating the associated costs.

Storage costs are incurred when goods are held in customs warehouses. Estimating this cost is straightforward since the daily unit cost for each warehouse is already known. Using the estimated release lead time, the total storage cost is calculated using the formula:

$$Storage\ Cost = Daily\ Unit\ Cost_{Warehouse} * Release\ LT \tag{3}$$

Transport costs are associated with moving goods from entry points to operational bases, a process managed by customs brokers. These costs are specified in contracts and depend on three key parameters: Entry Point, Destination Base, and Shipment Weight, which determines the type of vehicle required.

Broker handling costs are straightforward to determine as they are explicitly stated in the customs broker's contract. However, customs clearance costs are challenging to estimate due to their variability and dependence on customs regulations.

The total import cost is expressed as:

$$C_{total} = C_{transport}(X_i) + C_{storage}(Y_j) + C_{broker}(Z_k) + C_{customs} \tag{4}$$

where:

- X_i : *Chosen vehicle for transport*
- Y_j : *Entry point (Airport/Port)*
- Z_k : *customs broker assigned for clearance*

E. SOLUTION DEPLOYMENT

The deployment phase is the final stage in the CRISP-DM methodology, aiming to integrate the developed models into an operational environment for consistent use. This step ensures that the predictive models can function seamlessly in production without requiring frequent retraining or adjustments.

For deployment, the trained models are serialized into Pickle files (.pkl) using PyCaret. These files facilitate easy integration into other systems, as they can be reloaded without re-training. The selected best-performing models for each lead time are 3 Support Vector Machine (SVM) models for: CFD, Declaration, Transport Lead Time and Linear Regression model for Release Lead Time. Each model is saved as a separate Pickle file. For Transport Lead Time, a future enhancement could involve using Google Maps APIs or similar services for an initial estimate, which could then be fine-tuned by a simpler predictive model. These serialized files can be deployed on cloud platforms like AWS (Amazon Web Services) or Microsoft Azure using containerization services such as AWS ECS or Azure Container Apps. However, due to data confidentiality constraints, local deployment was selected as a viable solution until discussions with IT and information systems teams are finalized.

With the models exported in a usable format (pickle format), a complete end-to-end pipeline was developed. This pipeline enables users to input shipment details, perform necessary transformations, and calculate estimated lead times. It also provides scenarios for minimizing delays and costs. The process begins with users entering shipment details such as weight, business line, legal entity, quantity, and other relevant attributes. Users can also specify comparison dimensions, such as Mode of Transport (MOT), Customs Broker (CCA), or Port of Entry, and select multiple dimensions simultaneously for comparison. Based on the selected comparison dimensions, the system generates k instances of shipment data, where k represents the number of values for the chosen dimensions. For example, if the user wishes to compare lead times across three modes of transport, Sea Freight, Air Freight, and Land Freight, three data instances are created. These instances are then passed through the trained models, and the estimated lead times for each scenario are displayed. The workflow of this system is depicted in Figure 11

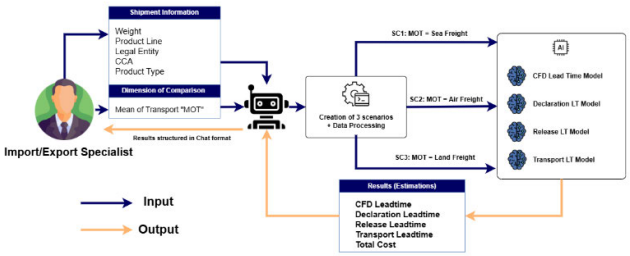

FIGURE 11. End-To-End pipeline for estimating lead times and costs.

The pipeline is flexible, allowing simultaneous comparisons across multiple dimensions such as: Mode of Transport (MOT): Air Freight, Land Freight, Sea Freight; Customs Broker (CCA): FD, VAT; Port of Entry: Alger-Port, Skikda-Port, HMD-Airport, etc.

To enhance usability, an interactive chatbot interface was developed. This interface simplifies interactions with the estimation system by allowing users to input shipment details and select comparison dimensions. The chatbot processes this information, sends it to the estimation system, and presents the results in a user-friendly manner. The chatbot is implemented using Python and the Streamlit library. It serves as the interface between the import-export specialist and the trained models, displaying estimated lead times and costs for each scenario. A working example of this chatbot is illustrated in Figure 12.

By integrating the trained models into an operational pipeline and providing an intuitive chatbot interface, this deployment approach ensures that the predictive models are not only accurate but also accessible and actionable for decision-makers in the import-export process.

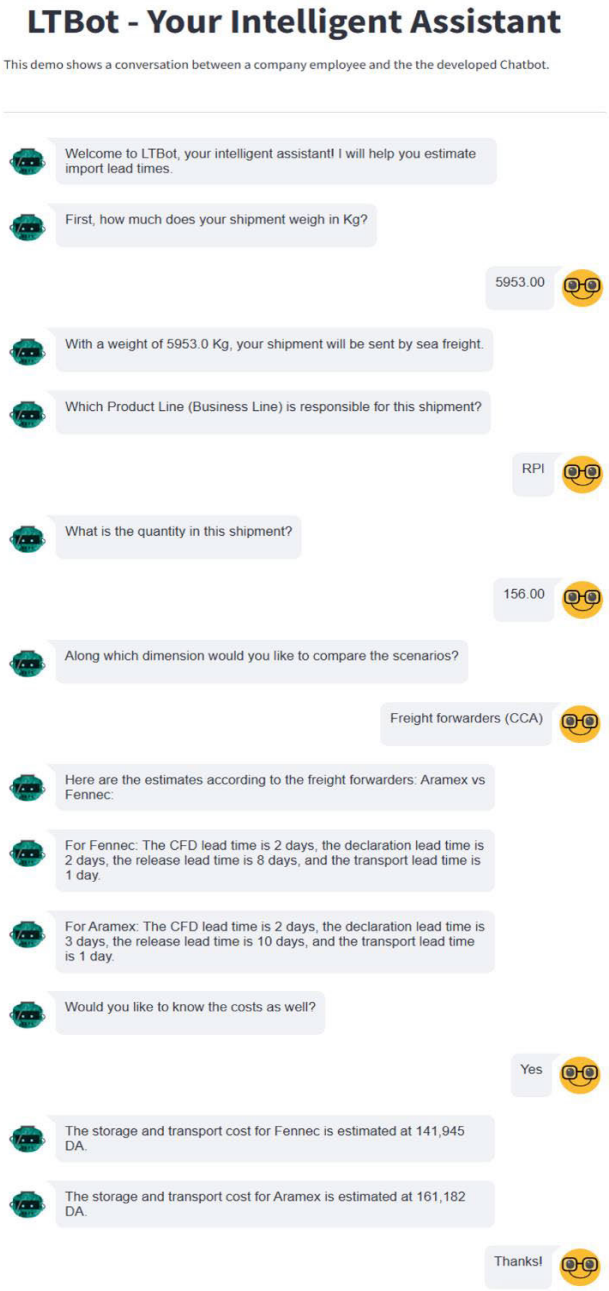

FIGURE 12. Lead estimation interface as an intelligent Chatbot.

V. DISCUSSION

From the results presented in Table 6, estimating lead times with ML based techniques can help greatly in getting more precise estimations compared to baseline guessing-based models used by different employees. By integrating Support Vector Machines (SVM) and Linear Regression models, we achieved significant reductions in forecasting errors across lead times, with improvements of up to 62% over baseline methods. Furthermore, the SVM model provides the best results in three different lead times compared to other models, while Linear Regression provides the lowest error for the release lead time, this could indicate the existence of a linear relationship between the release lead time and the provided variables.

These findings reinforce existing research on predictive analytics in supply chains, demonstrating that sophisticated data-driven models enhance process reliability, particularly in complex logistics environments [68], [72].

A. THEORETICAL IMPLICATIONS

From a theoretical perspective, this study advances the literature on data-driven supply chain management by highlighting the effectiveness of machine learning models in predicting import lead times. Unlike traditional forecasting approaches that rely on static averages, machine learning captures non-linear relationships, providing a dynamic solution. This is crucial in high-stakes industries like oil and gas, where accurate lead time predictions are vital for production continuity and cost efficiency, as discussed by Sharma et al. [73] or Ekram et al. [74]. Zouaghi [86] highlights how tacit knowledge can enhance decision-making, which, when integrated with machine learning, improves model adaptability.

Furthermore, our study underscores the value of hybrid data generation methods, combining Tabular Variational Autoencoder (TVAE) neural networks with discrete-event simulation. This approach addresses data limitations in sparse environments, aligning with recent theoretical advancements in the field [75], [76]. Zouaghi and Rouibi [87] further emphasize the importance of managing information distortion in supply chains, which adds another layer of complexity to lead time prediction. This hybrid approach enhances predictive accuracy in complex, data-limited environments.

B. METHODOLOGICAL IMPLICATIONS

Methodologically, this study demonstrates the effectiveness of combining machine learning with data augmentation techniques to enhance model performance, particularly in environments constrained by limited data availability. By leveraging the Tabular Variational Autoencoder (TVAE) as a synthetic data generator, we successfully expanded the dataset while preserving the original data's patterns and relationships. This approach significantly improved prediction reliability, supporting the growing consensus that well-generated synthetic data can mitigate overfitting and enhance the generalizability of machine learning models [77].

Furthermore, the integration of discrete-event simulation provided a process-specific understanding of lead times. This method allowed for the identification and modeling of bottlenecks and delay patterns within the logistics process, aligning with findings from simulation-based studies in supply chain management that emphasize its value for capturing process-level dynamics and variability [78].

The deployment of an interactive chatbot further highlights the practical application of technological tools to simplify complex machine learning outputs. By using Streamlit and Python, we developed a user-friendly interface that delivers real-time predictive insights, ensuring accessibility for end-users without technical expertise. This innovation aligns with recent advancements in supply chain digitalization, where intuitive real-time interfaces facilitate swift scenario analysis and enhance decision-making efficiency [79]. This methodological approach not only bridges the gap between technical complexity and operational usability but also underscores the role of digital tools in modernizing supply chain analytics.

C. STRATEGIC IMPLICATIONS

Model-driven decision-making offers significant strategic advantages in optimizing shipment scheduling. By leveraging predictive models, organizations can better forecast potential delays and optimize shipping routes, resulting in more accurate scheduling and timely deliveries. This ability to predict disruptions in advance enables proactive adjustments, reducing the risk of delays and improving overall operational performance.

In addition, integrating machine learning models into decision-making processes allows for effective cost optimization. Predictive insights help identify the most cost-effective shipping routes and minimize expenses such as storage fees and demurrage costs. By accurately predicting lead times and assessing transport options, companies can avoid inefficiencies and enhance budget control.

Furthermore, model-driven approaches are critical for selecting the right customs broker. By assessing brokers' past performance, machine learning models can identify the most reliable partners and predict the likelihood of customs delays, ensuring timely and compliant shipments. This reduces the risks associated with customs disruptions, safeguarding smooth operations.

The adaptability of model-driven decision-making also plays a key role in enhancing supply chain agility. Real-time data and continuous model refinement allow organizations to quickly respond to changes, such as regulatory shifts or unexpected disruptions, ensuring that operations remain on track.

Overall, model-driven decision-making enhances strategic decision-making by providing transparency and data-driven insights. This enables logistics managers to align decisions with broader business goals, optimizing resources, reducing lead times, and improving customer satisfaction, all of which contribute to a competitive advantage.

D. MANAGERIAL IMPLICATIONS

From a managerial perspective, this predictive system delivers significant operational advantages by enabling import-export specialists to make data-informed decisions based on scenario-driven lead time and cost estimations. These insights provide a strategic foundation for optimizing shipment scheduling, selecting efficient customs brokers, and determining the most reliable modes of transport, key factors for meeting internal KPIs and maintaining project timelines. The integration of model-driven decision-making highlights the strategic value of data analytics in streamlining logistics workflows, supporting previous findings that data-driven systems can substantially reduce lead times and associated costs in global supply chains [80].

Moreover, the system enhances managerial oversight by quantifying performance variations across multiple dimensions, such as customs broker efficiency, transport mode reliability, and port-specific delays. This level of granularity equips managers with actionable insights to identify and address process inefficiencies, allocate resources more effectively, and negotiate performance-based contracts with logistics partners. By integrating performance-based contracts with logistics partners, organizations can leverage system-generated insights to enhance accountability and service quality. These contracts, tied to key performance metrics such as on-time delivery and customs efficiency, encourage logistics partners to optimize their operations.

The data-driven approach reduces costs, mitigates risks by identifying potential disruptions early, and fosters continuous improvement. This strategic alignment enhances collaboration, ensuring that both parties work toward the same goal of improving overall supply chain performance. The chatbot interface, which facilitates scenario comparisons, further empowers managers to dynamically adjust logistics plans using real-time analytics, aligning with Li's observations [68] on agile supply chain management practices.

The system's scalable architecture, designed for potential deployment on cloud platforms like AWS or Microsoft Azure, reinforces its enterprise-wide applicability. This ensures that other divisions within the organization can replicate and benefit from similar predictive capabilities. As global supply chains face growing uncertainties, the system supports resilience-focused strategies, enabling organizations to adapt quickly to evolving trade conditions. Such flexibility is critical for enhancing supply chain robustness [72]. This approach not only drives operational efficiency but also strengthens the organization's capacity to respond proactively to disruptions in global logistics.

VI. CONCLUSION

This study demonstrates the transformative potential of machine learning and synthetic data generation for enhancing predictive accuracy in lead time and cost estimation within the import process. By implementing Support Vector Machines (SVM), Linear Regression, and Tabular Variational Autoencoders (TVAE), we successfully reduced forecasting errors and provided import-export specialists with actionable insights. The integration of these advanced analytics models with an interactive chatbot interface has proven effective in streamlining decision-making, facilitating scenario-based analysis, and empowering logistics teams with data-driven insights for optimizing resources. This approach highlights the importance of predictive analytics in refining supply chain management processes, particularly for time-sensitive and cost-intensive operations.

A. SYNTHESIS

The contributions of this study are primarily threefold. First, it demonstrates the power of machine learning in enhancing operational efficiency within logistics by accurately forecasting lead times and associated costs. By leveraging models trained on both real and synthetically augmented data, our approach enables organizations to better anticipate potential delays and strategically manage their resources. Second, the use of TVAE and discrete-event simulation illustrates an effective approach to overcoming data limitations often encountered in complex supply chain environments. These methods allow for the generation of high-quality synthetic data, providing a scalable solution for situations where historical data alone may be insufficient. Lastly, the development of a chatbot interface enhances accessibility and usability, translating complex analytics outputs into actionable insights directly usable by logistics personnel. This approach aligns with the growing emphasis on real-time, user-friendly systems in supply chain analytics, bridging the gap between technical outputs and practical applications.

B. LIMITATIONS

Despite the significant findings, this study does have several limitations. One key limitation is the dataset's time scope, restricted to the year 2022, which may not capture seasonal or year-to-year variations that impact import lead times. Additionally, while TVAE-generated data provides realistic augmentation, it may still lack some nuances inherent in real-world data, especially for highly variable customs clearance costs and unexpected regulatory changes. The study also relies on domain-specific constraints coded manually, which, while effective for this use case, may limit scalability across different products or regions. Lastly, while cloud deployment on platforms like AWS or Azure could improve scalability and accessibility, privacy and security concerns prevented a full-scale cloud deployment in this study, which could limit access for distributed teams and require local infrastructure.

C. FUTURE PERSPECTIVES

Building on the foundations of this work, future research could aim to further expand the dataset to capture multi-year trends and seasonal patterns, enhancing model accuracy and resilience. Integrating real-time data sources, such as GPS tracking, could refine transport lead time predictions and offer more responsive, dynamic insights. Exploring alternative synthetic data generation techniques like CTGAN could provide an even more comprehensive data augmentation approach, improving model diversity and handling more complex variable relationships.

To enhance the system's accessibility and scalability, future efforts could focus on cloud-based deployment with robust security protocols, allowing enterprise-wide access and supporting distributed logistics teams. Additionally, the implementation of adaptive learning could enable continuous model updates with incoming data, improving the system's ability to adjust to evolving conditions in the logistics environment. Finally, integrating reinforcement learning could empower the system to simulate and select optimal logistics strategies, enhancing its ability to proactively minimize lead times and costs.

In the same vein, to improve import lead time performance, a combined method proposed by Yen et al. [81] could be applied through integrating strategic, tactical, and operational decision-making into the data mining process, optimizing each phase to align with business objectives.

In conclusion, this study sets a foundation for predictive analytics in import logistics, showcasing how machine learning and synthetic data can drive efficiency and strategic advantages. By continuing to develop and expand this framework, organizations can enhance supply chain resilience, agility, and responsiveness to meet the dynamic demands of global trade.

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MOHAMED ANNIS SOUAMES received the M.S. degree in industrial management from the Ècole Nationale Polytechnique, Algiers, Algeria, and the joint M.Sc. degree in green networking and cloud computing from Leeds Beckett University, Leeds, U.K. He is currently a Software Engineer with Deutsche Telekom, Bonn, Germany. His research interests include distributed systems, cloud computing, and machine learning and their application in industrial environments, such as optimization of supply chains, computing resource scheduling, IoT communication, and distributed data systems.

LARBI ABDERRAHMANE MOHAMMEDI received the Diploma degree n industrial engineering from the École Nationale Polytechnique, Algiers, in 2022, and the master's degree in machine learning for data science from the Université Paris Cité. He is currently a Data Scientist with EssilorLuxottica, where he applies advanced analytics and machine learning techniques to optimize business processes. His research interests include supply chain management, predictive modeling, and operational optimization using data-driven methodologies.

ISKANDER ZOUAGHI received the M.Sc. degree from Paris-Dauphine University, and the Ph.D. degree from Grenoble-Alpes University. He is an Associate Professor (a Senior Lecturer), specializing in supply chain management, transactional and decision-based information systems, and data-oriented supply chain analytics. He is currently the Deputy Director of the École Nationale Polytechnique and an Adjunct Professor of business process management with the NEOMA Business School. His research focuses on the integration of advanced information systems and data-driven approaches to optimize decision-making in supply chain and business operations. He has contributed to several high-impact journals and international conferences, advancing the understanding of digital transformation; and member behavioral dynamics in supply chain management. He is actively engaged in academic and professional communities, collaborating with industry partners to apply research findings in real-world supply chain and information systems challenges.

ANGAPPA GUNASEKARAN received the Ph.D. degree in industrial engineering and operations research from Indian Institute of Technology Bombay. He is a Professor of supply chain management with the School of Business Administration. He was a Special Advisor for innovation and entrepreneurship (January 2024–May 2024) and the Director (August 2021–December 2023) of the School of Business Administration, Penn State Harrisburg. Prior to that, he held various academic leadership roles, including a Special Assistant to the Provost with California State University (CSU), Bakersfield, and the Dean of the School of Business and Public Administration, CSU Bakersfield. He was also the Dean of the Charlton College of Business and held leadership positions with the University of Massachusetts Dartmouth. He has held academic roles with several universities worldwide, including Brunel University, Monash University, and the University of Toronto. He has published over 400 articles, presented 50 papers, and spoken at conferences in over 20 countries. His research interests include benchmarking, agile manufacturing, management information systems, e-procurement, and supply chain management. He is on the editorial boards of more than 30 journals and is the editor of *Benchmarking*: An International Journal, and a North-American Editor of *International Journal of Enterprise Network Management*.

SAMIA BELDJOUDI is an Associate Professor with the Department of Industrial Engineering, École Nationale Polytechnique, Algiers. She leads the Data Science and Artificial Intelligence Team with the Advanced Multidisciplinary Industrial and Systems Engineering Laboratory. Her research focuses on data science, artificial intelligence, big data analytics, recommender systems, and knowledge graphs, with a particular emphasis on predictive maintenance and remaining useful life (RUL) estimation. She is actively involved in the development of AI-driven methodologies for data analysis and decision support in complex industrial systems.

ABDERRAZAK LAGHOUAG received the joint degree from the University of Algiers 3 and PierreMendès-France University, Grenoble 2. He is currently with the Department of Business Administration, College of Business, King Khalid University. He has guided several Ph.D. students. His research articles are published in high-impact factors journals. He is the Editor-in-Chief of *Journal of Money and Business*.